Results draft: Fernhill Software, version 2 of 3 (the trial designer's model)

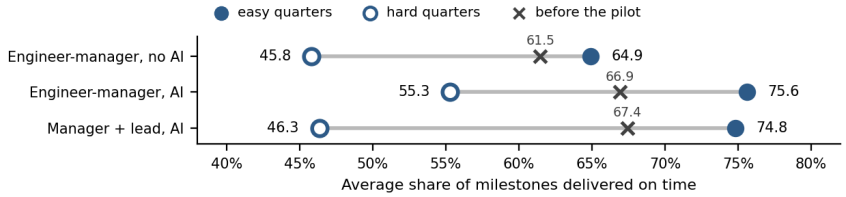
1. The question and the prediction

My question is causal. With AI held fixed, does the new structure raise delivery (the share of a team's committed milestones delivered by their dates) above engineer-managers', and what does AI alone add? The lot would let me call differences effects, but this draft answers only the descriptive question of how the arms compare and leaves the causal claim for Module 3. My model predicts AI adds about 8 points on easy quarters and little on hard ones, and the new structure shortens waits and adds about 5 points on hard quarters.

2. The charts

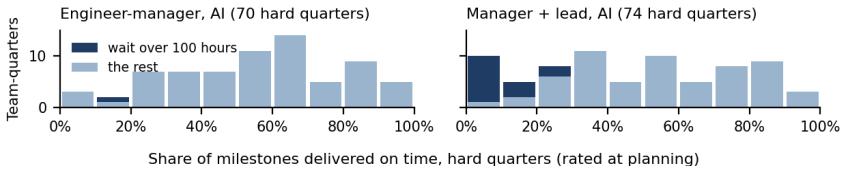
The first chart compares the three arms.

> **This chart answers:** how does delivery compare across the three arms on easy and hard quarters?
>
> Delivery is a measurement (a share), the arm is a label, and difficulty at planning is yes/no. Each dot is one arm's average over its pilot team-quarters of that kind (66 to 94), and each cross is its average over the eight quarters before the pilot (320).



The second chart looks at hard quarters in the two arms with AI.

> **This chart answers:** how are the hard quarters of the two arms with AI spread across delivery, and which of them had a wait over 100 hours?
>
> Each bar counts pilot team-quarters rated hard at planning, by delivered share. The dark part had a wait over 100 hours, and the light part is the rest.



3. The association

On AI alone, engineer-managers with AI delivered 10.7 points more on easy quarters and 9.5 more on hard ones than those without. This is measured at the team-quarter level, over 320 pilot team-quarters from 80 teams, and it compares teams offered AI, not teams that used it. But the lot was unlucky. Before the pilot, the no-AI arm already delivered 61.5% against 66.9%, so against each arm's own before-period the gap is 5.2 points on easy quarters and 4.1 on hard.

On the structure, with AI in both arms, the new structure delivered 0.8 points less on easy quarters and 8.9 less on hard ones (9.4 against the before-periods), over 320 team-quarters from 80 teams. It holds by team (0.2 points down under engineer-managers with AI, 5.8 in the new structure) and without the first

pilot quarter (7.3 points less on hard quarters). **The average and the median disagree on delivery.** The new structure's pilot median is 71.4%, level with engineer-managers with AI and 1.6 points above its own before-period, yet its average is 61.7% against 66.7%. The weak quarters pull the average down, and the median team still fell 5.7 points.

On hard quarters the new structure's median wait was shorter (24.6 hours against 27.8, with 2 blank waits left out), though its average was longer (83.0 against 29.4). People-skill ratings were higher under professional managers (0.78 against 0.69, 33 blank left out).

And the new structure's hard quarters split. As many delivered 70% or more as under engineer-managers with AI (20 against 19), but nearly twice as many delivered under 30% (23 against 12). The split follows the waits. Of its 73 hard quarters with a recorded wait, 14 waited over 100 hours (1 of 69 under engineer-managers with AI) and delivered 8.8% on average. The other 59 delivered 55.4%, level with engineer-managers with AI (55.3%).

4. Against the model

The Brief named three ways the model would be wrong, and two hold. AI's gain is nearly as large on hard quarters as easy (9.5 points against 10.7, and 4.1 against 5.2 after the before-periods), so the first holds. The new structure's median wait on hard quarters is shorter, so the second doesn't hold, but its average is longer. At the median its waits are shorter yet its delivery is 8.9 points lower, so the third holds.

But the median hides something. The shortfall sits in the 14 hard quarters with a wait over 100 hours, and the rest delivered as engineer-managers with AI did, so the queue may limit delivery after all.

5. What this does not establish

The strongest other explanations are the three that fit a hard-quarter shortfall, and the file can test two of them. If leads were overloaded, I would expect backlogs in the new structure. Escalations were still open at quarter's close in 1% of its team-quarters, against 34% to 42% under engineer-managers, so backlogs don't fit, though its long waits do.

If AI misled teams on hard work, I would expect every AI arm to suffer there, but engineer-managers with AI did better than those without (55.3% against 45.8%).

My Data & Methods concern, misjudged difficulty, shrinks hard–easy differences, so the shortfall is, if anything, understated. Noticed, not pursued. Leads left 11.2% of new-structure teams a quarter, against 3% to 4% elsewhere, and all 18 who left had been the team's manager.

The decision needs a model that explains, because the average hides a split, and only knowing what splits it says where the structure would work.

What would settle it is a better diagram, not more of the same data. The shortfall belongs to the new structure on hard work, and within it to quarters whose waits explode, not to AI. Every escalation there passes between the lead, who judges the technical path, and the manager, who sets priorities, and our diagram has no box for that. Module 3 draws the diagram, and that box is where to look first, measured from the first month.

Note on AI use

I used Gemini (Willamette account) for the comparisons and charts. Asked early, it judged the wait prediction contradicted from the all-quarter average wait. I added each arm's before-period, split the waits by difficulty, used the median, and kept the long waits it called outliers.